

Second Partial

Fco. Vazquez

Tec de Monterrey

Chain rule

Definition of the chain rule

The chain rule allow us to compute the derivative of a composition of functions $f(x) = (h \circ g)(x) = h(g(x))$. The rule goes as follows,

Leibniz notation

$$\frac{df(x)}{dx} = \left[\frac{dh(x)}{dx} \circ g(x) \right] \frac{dg(x)}{dx}$$

Lagrange notation

$$f'(x) = h'(g(x)) \cdot g'(x)$$

Chain rule

Examples

Let's compute the derivative of $f(x) = \ln(\cos(x))$.

$$\begin{aligned} f'(x) &= \ln'(\cos(x)) \cdot \cos'(x) \\ &= \frac{1}{\cos(x)} \cdot -\sin(x) \\ &= -\tan(x) \end{aligned}$$

Chain rule

Examples

Let's compute the derivative of $f(x) = \cos(2\pi x - \pi)$.

$$\begin{aligned} f'(x) &= \cos'(2\pi x - \pi) \cdot (2\pi x - \pi)' \\ &= -\sin(2\pi x - \pi) \cdot (2\pi) \\ &= -2\pi \sin(2\pi x - \pi) \end{aligned}$$

Chain rule

Examples

Let's compute the derivative of $f(x) = e^{-\frac{x^2}{4}}$.

$$\begin{aligned} f'(x) &= \left[\frac{de^x}{dx} \circ -\frac{x^2}{4} \right] \cdot \left(-\frac{x^2}{4} \right)' \\ &= e^{-\frac{x^2}{4}} \cdot \left(-\frac{x}{2} \right) \\ &= -\frac{x}{2} e^{-\frac{x^2}{4}} \end{aligned}$$

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Class Activity

Compute the derivative of the following functions using the chain rule:

$$f(x) = \ln(e^x), \quad g(x) = e^{\cos(x)}$$